

# How the discharge app protects patient data before any AI processing

A visual, synthetic-data report showing the current app redactor, the privacy test suite results, and before/after examples for standard and adversarial discharge paperwork.

App release 1.1.2

4400/4400

standard PHI literals redacted

530/530

adversarial PHI literals blocked

0

documents with PHI reaching  
Groq in the egress test

Generated 2026-05-14T17:57:20.025Z. All patient data shown here is synthetic test data, not real patient data.

MyDischarge is designed so raw patient-identifying data is removed before language-model processing. The tests below verify the current app redactor against synthetic discharge paperwork with canary tokens and known PHI fields.

- **Photos stay on device.** OCR reads discharge paperwork locally; image files are not uploaded for parsing.
- **Patient identifiers are redacted on device.** The app removes names, dates of birth, ages, SSNs, patient phone numbers, patient email addresses, MRNs, insurance IDs, account numbers, and home addresses.
- **Clinical utility is preserved where possible.** Diagnosis, medications, doctor names, hospital names, hospital phone numbers, and follow-up dates are intentionally preserved when they are not patient identifiers.
- **Network requests are de-identified.** The egress tests simulate what would be sent after client redaction and the server-side safety barrier.
- **No analytics or tracking SDKs are used in the app.** Local session data is stored on-device in encrypted storage.

## Standard Multilingual Corpus

The standard corpus contains 440 synthetic documents across 15 languages with 4400 labeled PHI instances.

| PHI class | Instances tested | Redacted | Leaked | Redaction rate |
|-----------|------------------|----------|--------|----------------|
| name      | 440              | 440      | 0      | 100.00%        |
| dob       | 440              | 440      | 0      | 100.00%        |
| age       | 440              | 440      | 0      | 100.00%        |
| ssn       | 440              | 440      | 0      | 100.00%        |
| phone     | 440              | 440      | 0      | 100.00%        |
| email     | 440              | 440      | 0      | 100.00%        |
| mrn       | 440              | 440      | 0      | 100.00%        |
| insurance | 440              | 440      | 0      | 100.00%        |
| account   | 440              | 440      | 0      | 100.00%        |
| address   | 440              | 440      | 0      | 100.00%        |

| Language | PHI instances | Redacted | Leaked | Redaction rate |
|----------|---------------|----------|--------|----------------|
| EN       | 600           | 600      | 0      | 100.00%        |
| ES       | 600           | 600      | 0      | 100.00%        |
| ZH       | 300           | 300      | 0      | 100.00%        |
| VI       | 300           | 300      | 0      | 100.00%        |
| KO       | 200           | 200      | 0      | 100.00%        |
| TL       | 200           | 200      | 0      | 100.00%        |
| AR       | 300           | 300      | 0      | 100.00%        |
| FR       | 300           | 300      | 0      | 100.00%        |
| RU       | 200           | 200      | 0      | 100.00%        |
| PT       | 300           | 300      | 0      | 100.00%        |
| HT       | 200           | 200      | 0      | 100.00%        |
| DE       | 300           | 300      | 0      | 100.00%        |
| JA       | 200           | 200      | 0      | 100.00%        |
| HI       | 200           | 200      | 0      | 100.00%        |
| FA       | 200           | 200      | 0      | 100.00%        |

## Provider Information Preservation

Provider details are medically important, so the test suite measures whether they are preserved while patient identifiers are removed.

| Provider detail  | Kept in redacted output |
|------------------|-------------------------|
| hospital name    | 92.7%                   |
| doctor name      | 100.0%                  |
| medication       | 100.0%                  |
| diagnosis        | 100.0%                  |
| appointment date | 90.9%                   |
| hospital phone   | 100.0%                  |

## Adversarial and Stress Tests

The stress corpus includes OCR confusions, narrative text, typos, multiline labels, duplicated identifiers, and unlabeled identifiers.

| Category   | Docs | PHI instances | Literal leaks at Groq input | Canary leaks | Docs with any leak |
|------------|------|---------------|-----------------------------|--------------|--------------------|
| ocr        | 10   | 100           | 0                           | 0            | 0                  |
| narrative  | 10   | 100           | 0                           | 0            | 0                  |
| typos      | 10   | 100           | 0                           | 0            | 0                  |
| multiline  | 10   | 100           | 0                           | 0            | 0                  |
| duplicated | 10   | 30            | 0                           | 0            | 0                  |
| unlabeled  | 10   | 100           | 0                           | 0            | 0                  |

## Egress Pipeline Test

This test simulates each stage from raw input through the text that would be sent to the LLM.

| Pipeline stage         | Canary leaks | PHI literal leaks | Docs with any leak |
|------------------------|--------------|-------------------|--------------------|
| raw_input              | 440          | 4050              | 440                |
| after_client_redaction | 0            | 0                 | 0                  |
| after_server_sanitize  | 0            | 0                 | 0                  |
| sent_to_groq           | 0            | 0                 | 0                  |

# Before and After Redaction Examples

The following pages show synthetic source text with PHI highlighted, the current app redaction output, and the status of each labeled test value.

## Standard corpus example

English synthetic discharge instruction, representative of the 440-document multilingual corpus.

### Original synthetic discharge text with PHI highlighted

EMERGENCY DEPARTMENT DISCHARGE INSTRUCTIONS  
Ben Taub General Hospital  
Hospital Phone: 713-500-7878

Patient Name: **Octavio Mogensen-ZQK0001**  
DOB: **05/02/1931**    Age: **51**  
SSN: **900-11-0001**  
Patient Phone: **555-010-0001**  
Patient Email: **canary-zqk0001@example.invalid**  
Patient Address: **2 Canary-ZQK0001 Lane, Testville, TX 77000**  
MRN: **MRNZQK0001**  
Insurance ID: **INSZQK0001XYZ9**  
Account #: **90000001**

Attending: Dr. Luber  
Diagnosis: community-acquired pneumonia  
Medication: Take metoprolol 25 mg as directed until finished.  
Follow-up: Please schedule a follow-up appointment on 5/24/2026 with your primary care provider.  
If you develop chest pain, shortness of breath, or fever above 101°F, call 911 or return to the ED immediately.

### Current app redaction output

EMERGENCY DEPARTMENT DISCHARGE **[REDACTED]**  
Ben Taub General Hospital  
Hospital **[REDACTED]** : 713-500-7878

**[REDACTED]** : **[REDACTED]**  
DOB: **[REDACTED]**    Age: **[REDACTED]** SSN: **[REDACTED]**  
**[REDACTED]** **[REDACTED]** : **[REDACTED]**  
**[REDACTED]** **[REDACTED]** : **[REDACTED]**  
**[REDACTED]**  
**[REDACTED]**  
**[REDACTED]** : **[REDACTED]**  
Account #: **[REDACTED]**

Attending: Dr. Luber  
Diagnosis: community-acquired pneumonia  
Medication: Take metoprolol 25 mg as directed until finished.  
Follow-up: Please schedule a follow-up appointment on 5/24/2026 with your primary care provider.  
If you develop chest pain, shortness of breath, or fever above 101°F, call 911 or return to the ED immediately.

| PHI class | Synthetic value tested                     | Result   |
|-----------|--------------------------------------------|----------|
| name      | Octavio Mogensen-ZQK0001                   | Redacted |
| dob       | 05/02/1931                                 | Redacted |
| age       | 51                                         | Redacted |
| ssn       | 900-11-0001                                | Redacted |
| phone     | 555-010-0001                               | Redacted |
| email     | canary-zqk0001@example.invalid             | Redacted |
| mrn       | MRNZQK0001                                 | Redacted |
| insurance | INSZQK0001XYZ9                             | Redacted |
| account   | 90000001                                   | Redacted |
| address   | 2 Canary-ZQK0001 Lane, Testville, TX 77000 | Redacted |

# Adversarial ocr example

One synthetic stress-test document from the ocr category.

## Original synthetic discharge text with PHI highlighted

DISCHARGE INSTRUCTIONS  
Facility: Memorial Hermann TMC  
Hospital Phone: 713-500-7878

Patient Name: Marcus Thomton-ZQK1001  
D0B: 03/14/1962 Age: 63  
SSN: 900-44-1001  
Patient Phone: 555-020-1001  
Patient Email: canary-zqk1001@example.invalid  
Patient Address: 14 Canary-ZQK1001 Ct, Testville, TX 77000  
MRN: MRNZQK1001  
Insurance ID: INSZQK1001XYZ  
Account: 90001001

Follow up in 1 week.

## Current app redaction output

DISCHARGE [REDACTED]  
Facility: Memorial Hermann TMC  
Hospital [REDACTED] : 713-500-7878

[REDACTED] : [REDACTED]  
D0B: [REDACTED] Age: [REDACTED] SSN: [REDACTED]  
[REDACTED] [REDACTED] : [REDACTED]  
[REDACTED] [REDACTED] : [REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED] : [REDACTED]  
Account: [REDACTED]

Follow up in 1 week.

| PHI class | Synthetic value tested                    | Result                 |
|-----------|-------------------------------------------|------------------------|
| name      | Marcus Thornton-ZQK1001                   | Source variant handled |
| dob       | 03/14/1962                                | Redacted               |
| age       | 63                                        | Redacted               |
| ssn       | 900-44-1001                               | Redacted               |
| phone     | 555-020-1001                              | Redacted               |
| email     | canary-zqk1001@example.invalid            | Redacted               |
| mrn       | MRNZQK1001                                | Redacted               |
| insurance | INSZQK1001XYZ                             | Redacted               |
| account   | 90001001                                  | Redacted               |
| address   | 14 Canary-ZQK1001 Ct, Testville, TX 77000 | Redacted               |

# Adversarial narrative example

One synthetic stress-test document from the narrative category.

## Original synthetic discharge text with PHI highlighted

EMERGENCY DEPARTMENT NOTE  
Patient **Marcus Thornton-ZQK1011** presented today complaining of sore throat.  
Date of birth **03/14/1962**; the patient is **63** years old.  
His record number is **MRNZQK1011** and his insurance plan ID is **INSZQK1011XYZ**.  
Reachable at **555-020-1011** or **canary-zqk1011@example.invalid**.  
Lives at **14 Canary-ZQK1011 Ct, Testville, TX 77000**.  
**Marcus Thornton-ZQK1011** was discharged home in stable condition.  
Social security on file ends in 1011 (full: **900-44-1011**).  
Account **90001011**.  
If Marcus has any questions, please call our clinic at 713-500-7878.

## Current app redaction output

EMERGENCY DEPARTMENT NOTE  
[REDACTED] [REDACTED] g of sore throat.  
Date of birth [REDACTED]; the [REDACTED] is [REDACTED] years old.  
His record number is [REDACTED] and his insurance plan ID is [REDACTED].  
Reachable at [REDACTED] or [REDACTED].  
[REDACTED]  
[REDACTED] [REDACTED] was discharged home in stable condition.  
Social security on file ends in 1011 (full: [REDACTED]).  
Account [REDACTED].  
If [REDACTED] has any questions, please call our clinic at 713-500-7878.

| PHI class | Synthetic value tested                    | Result   |
|-----------|-------------------------------------------|----------|
| name      | Marcus Thornton-ZQK1011                   | Redacted |
| dob       | 03/14/1962                                | Redacted |
| age       | 63                                        | Redacted |
| ssn       | 900-44-1011                               | Redacted |
| phone     | 555-020-1011                              | Redacted |
| email     | canary-zqk1011@example.invalid            | Redacted |
| mrn       | MRNZQK1011                                | Redacted |
| insurance | INSZQK1011XYZ                             | Redacted |
| account   | 90001011                                  | Redacted |
| address   | 14 Canary-ZQK1011 Ct, Testville, TX 77000 | Redacted |

# Adversarial typos example

One synthetic stress-test document from the typos category.

## Original synthetic discharge text with PHI highlighted

DISCHARGE SUMMARY  
Patient Name: **Marcus Thornton-ZQK1021**  
D.O.B: **03/14/1962**  
Home Address: **14 Canary-ZQK1021 Ct, Testville, TX 77000**  
Pt Phone: **555-020-1021**  
SSN: **900-44-1021**  
MRN: **MRNZQK1021**  
Insurance: **INSZQK1021XYZ**  
Acct: **90001021**  
Email: **canary-zqk1021@example.invalid**

Age: **63** y  
Attending: Dr. Lee  
Dx: viral pharyngitis  
Rx: supportive care

## Current app redaction output

[REDACTED]  
[REDACTED] : [REDACTED]  
D.O.B: [REDACTED]  
[REDACTED]  
[REDACTED] : [REDACTED]  
SSN: [REDACTED]  
[REDACTED]  
Insurance: [REDACTED]  
Acct: [REDACTED]  
Email: [REDACTED]

Age: [REDACTED]  
Attending: Dr. Lee  
Dx: viral pharyngitis  
Rx: supportive care

| PHI class | Synthetic value tested                    | Result   |
|-----------|-------------------------------------------|----------|
| name      | Marcus Thornton-ZQK1021                   | Redacted |
| dob       | 03/14/1962                                | Redacted |
| age       | 63                                        | Redacted |
| ssn       | 900-44-1021                               | Redacted |
| phone     | 555-020-1021                              | Redacted |
| email     | canary-zqk1021@example.invalid            | Redacted |
| mrn       | MRNZQK1021                                | Redacted |
| insurance | INSZQK1021XYZ                             | Redacted |
| account   | 90001021                                  | Redacted |
| address   | 14 Canary-ZQK1021 Ct, Testville, TX 77000 | Redacted |

# Adversarial multiline example

One synthetic stress-test document from the multiline category.

## Original synthetic discharge text with PHI highlighted

After Visit Summary

Patient Name:  
Marcus Thornton-ZQK1031

Date of Birth:  
03/14/1962

Home Address:  
14 Canary-ZQK1031 Ct  
Testville, TX 77000

Contact:  
Cell: 555-020-1031  
Email: canary-zqk1031@example.invalid

Medical Record Number: MRNZQK1031  
Insurance Member ID: INSZQK1031XYZ  
SSN: 900-44-1031  
Account #: 90001031

Age:  
63

Attending physician: Dr. Carlos Ruiz  
Diagnosis: urinary tract infection  
Medication: nitrofurantoin 100 mg twice daily for 5 days.

## Current app redaction output

[REDACTED]

[REDACTED] :  
[REDACTED]

Date of Birth:  
[REDACTED]

[REDACTED] :  
[REDACTED]  
[REDACTED]

Contact:  
Cell: [REDACTED]  
Email: [REDACTED]

[REDACTED] : [REDACTED]  
[REDACTED] : [REDACTED]  
SSN: [REDACTED]  
Account #: [REDACTED]

Age:  
[REDACTED]

Attending physician: Dr. Carlos Ruiz  
Diagnosis: urinary tract infection  
Medication: nitrofurantoin 100 mg twice daily for 5 days.

| PHI class | Synthetic value tested                    | Result                 |
|-----------|-------------------------------------------|------------------------|
| name      | Marcus Thornton-ZQK1031                   | Redacted               |
| dob       | 03/14/1962                                | Redacted               |
| age       | 63                                        | Redacted               |
| ssn       | 900-44-1031                               | Redacted               |
| phone     | 555-020-1031                              | Redacted               |
| email     | canary-zqk1031@example.invalid            | Redacted               |
| mrn       | MRNZQK1031                                | Redacted               |
| insurance | INSZQK1031XYZ                             | Redacted               |
| account   | 90001031                                  | Redacted               |
| address   | 14 Canary-ZQK1031 Ct, Testville, TX 77000 | Source variant handled |

# Adversarial duplicated example

One synthetic stress-test document from the duplicated category.

## Original synthetic discharge text with PHI highlighted

Memorial Hermann TMC – Discharge Summary for **Marcus Thornton-ZQK1041**

Patient Name: **Marcus Thornton-ZQK1041**  
DOB: **03/14/1962**  
MRN: **MRNZQK1041**

Dear Marcus,

You were seen today in our emergency department. Marcus, you had symptoms of chest pain. We performed an ECG and labs. **Marcus Thornton-ZQK1041**, your results were normal.

Marcus, please follow up with your primary care provider in 3 days.

If you, Marcus, develop any new symptoms, return to the ED.

Patient signature: **Marcus Thornton-ZQK1041**      Date: 4/13/2026  
Printed: **Marcus Thornton-ZQK1041**  
Copy to file under MRN **MRNZQK1041**.

## Current app redaction output

[REDACTED] – [REDACTED] for [REDACTED]

[REDACTED] : [REDACTED]  
DOB: [REDACTED]  
[REDACTED]

[REDACTED] [REDACTED] ,

You were seen today in our emergency department. [REDACTED] , you had symptoms of chest pain. We performed an ECG and labs. [REDACTED] , your results were normal.

[REDACTED] , please follow up with your primary care provider in 3 days.

If you, [REDACTED] , develop any new symptoms, return to the ED.

[REDACTED] signature: [REDACTED]      [REDACTED] : 4/13/2026  
Printed: [REDACTED]  
Copy to file under [REDACTED] .

| PHI class | Synthetic value tested  | Result   |
|-----------|-------------------------|----------|
| name      | Marcus Thornton-ZQK1041 | Redacted |
| dob       | 03/14/1962              | Redacted |
| mrn       | MRNZQK1041              | Redacted |

# Adversarial unlabeled example

One synthetic stress-test document from the unlabeled category.

## Original synthetic discharge text with PHI highlighted

Visit Summary

Marcus Thornton-ZQK1051  
03/14/1962  
14 Canary-ZQK1051 Ct, Testville, TX 77000  
555-020-1051  
canary-zqk1051@example.invalid  
MRNZQK1051  
INSZQK1051XYZ  
900-44-1051  
90001051

Age 63  
Reason for visit: acute bronchitis.  
Plan: rest, fluids, follow up if not improving.

## Current app redaction output

[REDACTED]

[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]  
[REDACTED]

Age [REDACTED] Reason for [REDACTED]: acute bronchitis.  
Plan: rest, fluids, follow up if not improving.

| PHI class | Synthetic value tested                    | Result   |
|-----------|-------------------------------------------|----------|
| name      | Marcus Thornton-ZQK1051                   | Redacted |
| dob       | 03/14/1962                                | Redacted |
| age       | 63                                        | Redacted |
| ssn       | 900-44-1051                               | Redacted |
| phone     | 555-020-1051                              | Redacted |
| email     | canary-zqk1051@example.invalid            | Redacted |
| mrn       | MRNZQK1051                                | Redacted |
| insurance | INSZQK1051XYZ                             | Redacted |
| account   | 90001051                                  | Redacted |
| address   | 14 Canary-ZQK1051 Ct, Testville, TX 77000 | Redacted |